PRAKTIKUM SISTEM OPERASI

MODUL IV

MEMBACA SOURCE CODE XINU

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

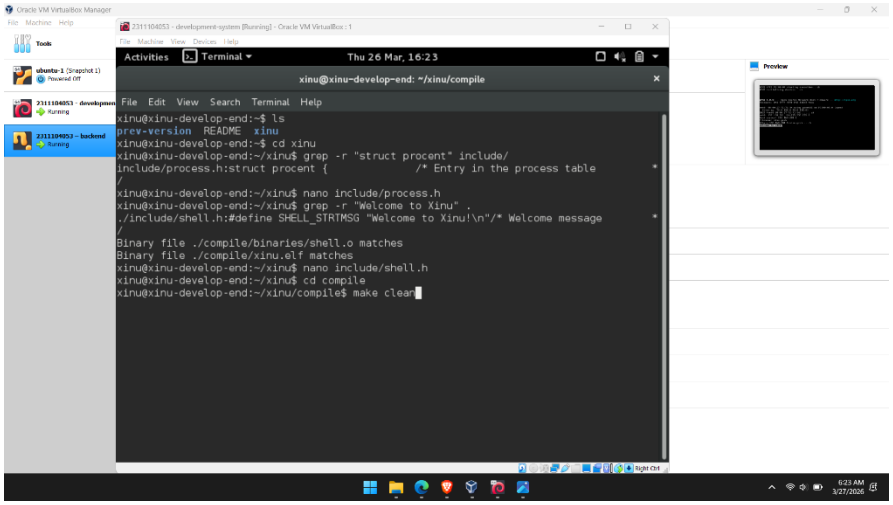
DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

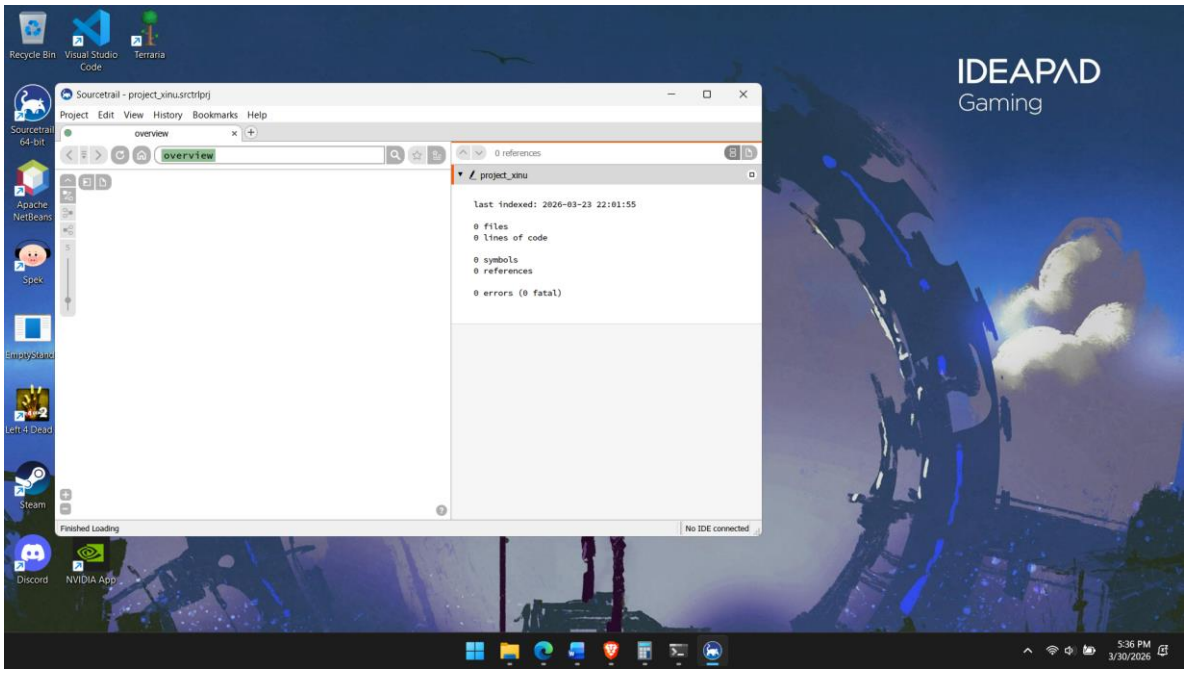
JURNAL

1. **Nama image yang dihasilkan setelah melakukan kompilasi pada Xinu**
   Jawaban:



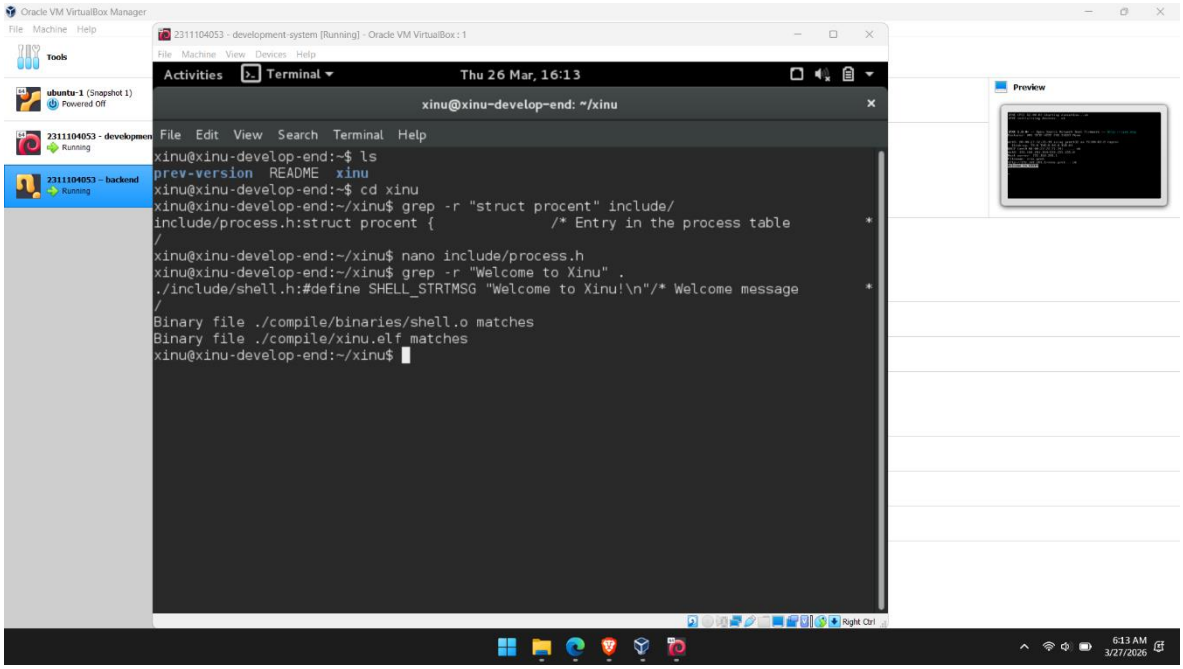
   Nama Image: xinu.boot.
   Ukuran: 159180 bytes.
   Lokasi: Folder xinu/compile

2. **Membaca source code Xinu**



3. **Struktur Data Proses**

Jawaban:

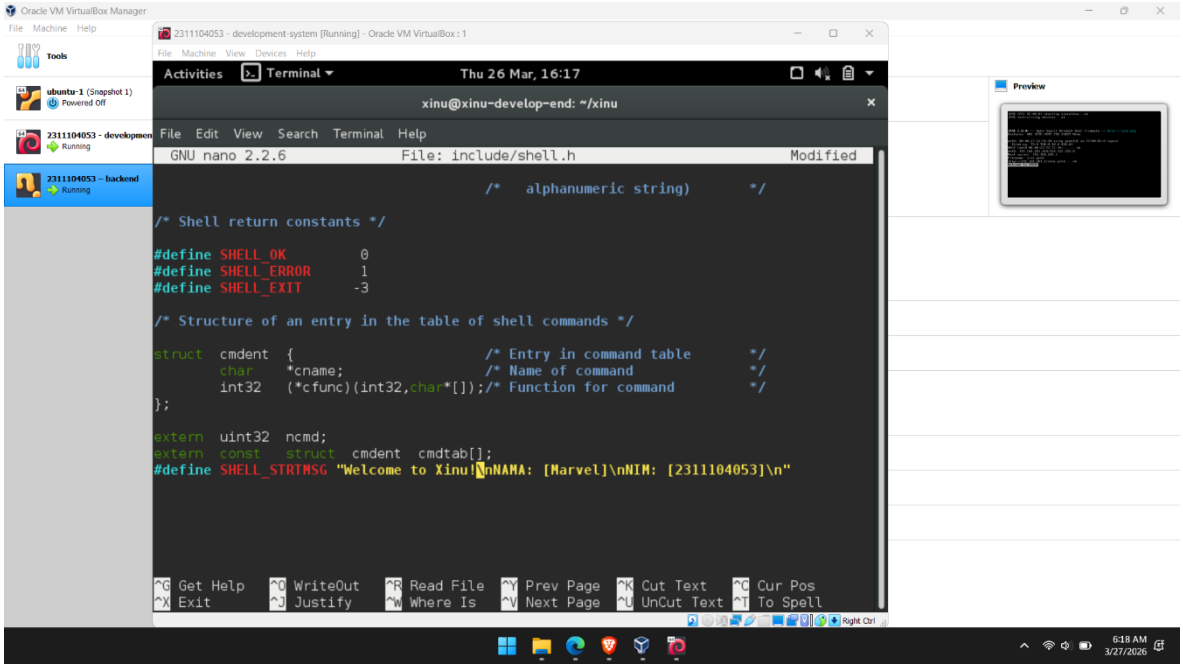


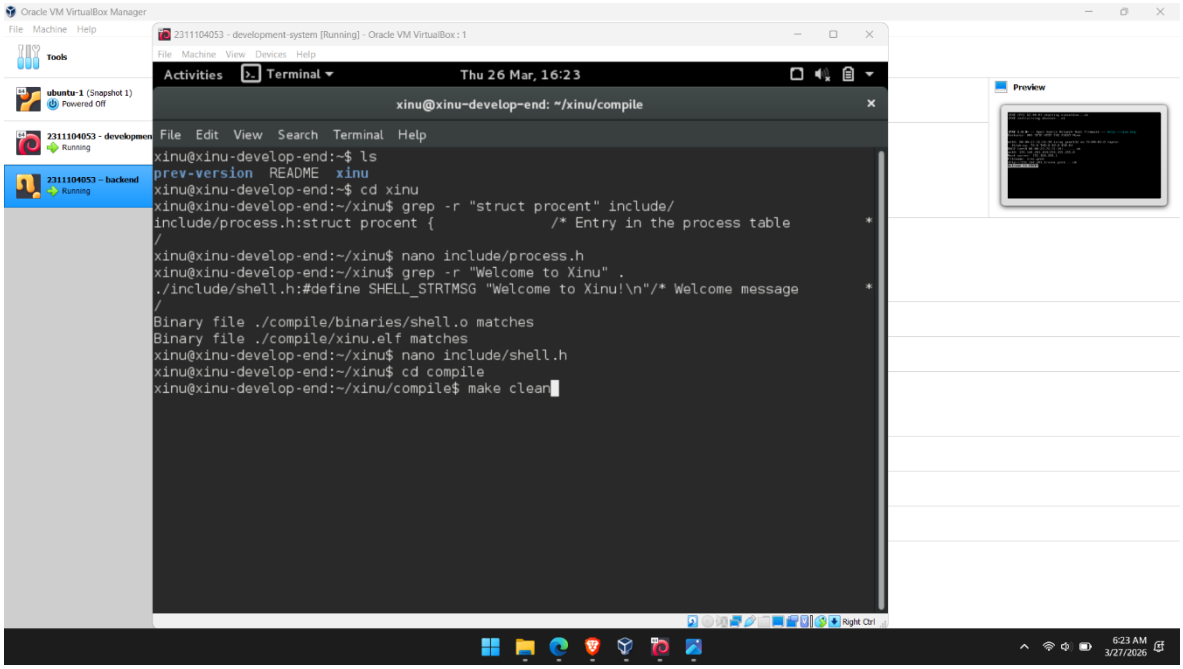
File Struktur Data: include/process.h.

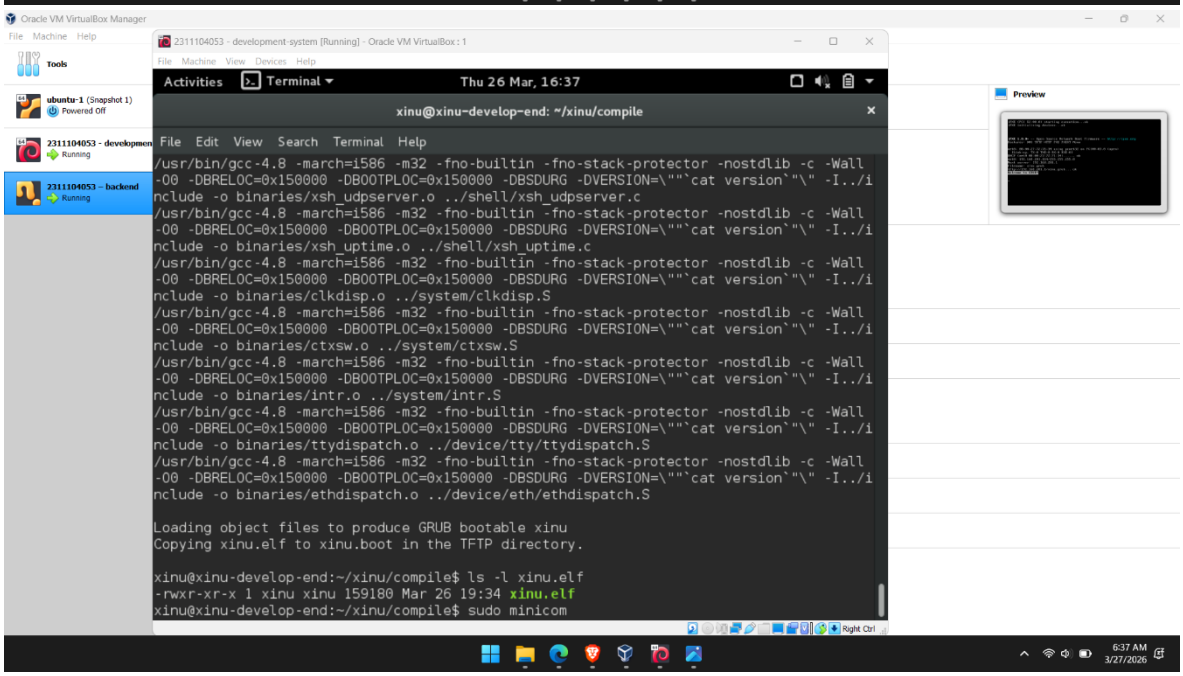
Informasi yang Disimpan: Di dalam struct procent, informasi yang disimpan meliputi status proses (prstate), prioritas proses (prprio), stack pointer (prstkptr), basis stack (prstkbase), panjang stack (prstklen), dan nama proses (prname)

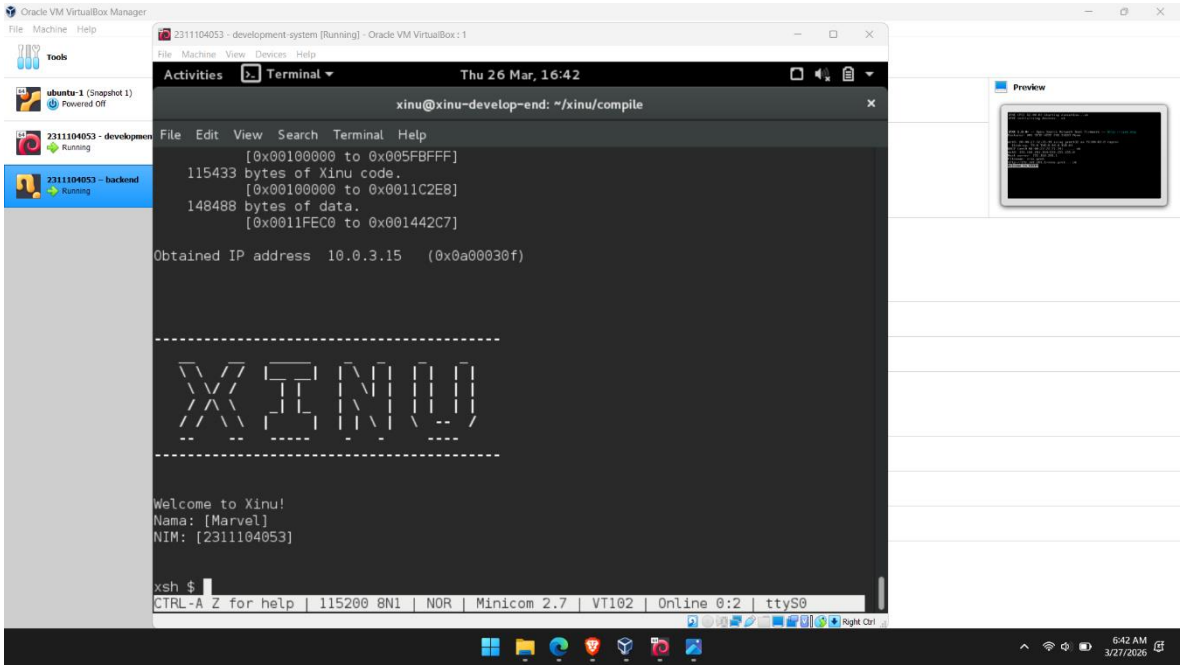
4. Modifikasi Banner

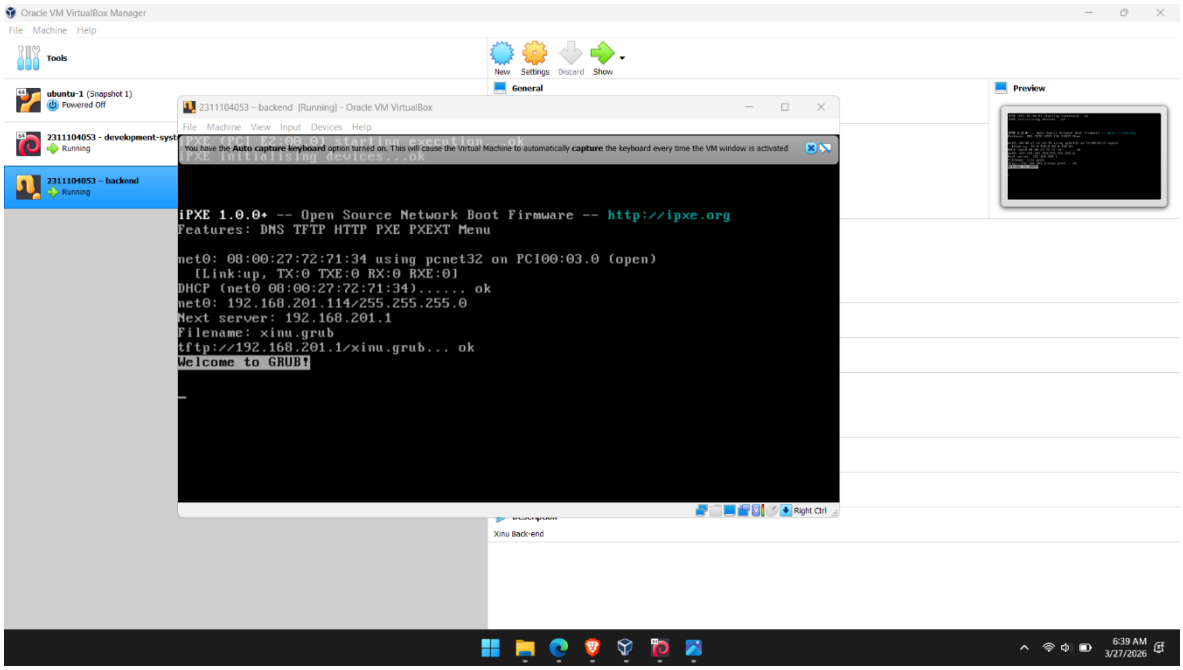
Jawaban:




Oracle VM VirtualBox Manager
File Machine Help
Tools
ubuntu-1 (Snapshot 1)
Powered Off
2311104053 - developmen
Running
2311104053 - backend
Running
2311104053 - development-system [Running] - Oracle VM VirtualBox : 1
File Machine View Devices Help
Activities
Terminal
Thu 26 Mar, 16:23
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ ls
prev-version README xinu
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ grep -r "struct procent" include/
include/process.h:struct procent {   /* Entry in the process table   *
/
xinu@xinu-develop-end:~/xinu$ nano include/process.h
xinu@xinu-develop-end:~/xinu$ grep -r "Welcome to Xinu" .
./include/shell.h:#define SHELL_STRTMSG "Welcome to Xinu!\n"/* Welcome message   *
/
Binary file ./compile/binaries/shell.o matches
Binary file ./compile/xinu.elf matches
xinu@xinu-develop-end:~/xinu$ nano include/shell.h
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make clean
Preview
Right Ctrl
6:23 AM
3/27/2026


Oracle VM VirtualBox Manager
File Machine Help
Tools
ubuntu-1 (Snapshot 1)
Powered Off
2311104053 - developmen
Running
2311104053 - backend
Running
2311104053 - development-system [Running] - Oracle VM VirtualBox : 1
File Machine View Devices Help
Activities
Terminal
Thu 26 Mar, 16:37
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/xsh_udpserver.o ../shell/xsh_udpserver.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/clkdisp.o ../system/clkdisp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-O0 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\""`cat version`"\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S
Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.
xinu@xinu-develop-end:~/xinu/compile$ ls -l xinu.elf
-rwxr-xr-x 1 xinu xinu 159180 Mar 26 19:34 xinu.elf
xinu@xinu-develop-end:~/xinu/compile$ sudo minicom
Preview
Right Ctrl
6:37 AM
3/27/2026





a. **File yang menyimpan banner Xinu:**
   "File yang menyimpan definisi teks banner Xinu terletak pada direktori xinu/include/shell.h. Di dalam file ini, saya memodifikasi variabel SHELL_BAN untuk mengubah tampilan logo dan SHELL_STRTMSG untuk menambahkan identitas Nama dan NIM."

b. **File yang menampilkan banner Xinu:**

"File yang bertugas untuk memanggil dan menampilkan banner tersebut ke layar adalah xinu/shell/shell.c. File berekstensi .c ini menggunakan definisi pesan dari file header untuk dicetak saat shell pertama kali dijalankan."

c. **& d. Proses Modifikasi dan Kompilasi:**
"Langkah-langkah yang saya lakukan adalah sebagai berikut:
1. Memberikan identitas Nama dan NIM dalam bentuk comment di dalam source code sesuai aturan praktikum.
2. Mengubah isi welcome banner sesuai selera dengan menyisipkan Nama: Marvel dan NIM: 2311104053.
3. Melakukan kompilasi ulang di terminal Linux dengan masuk ke direktori xinu/compile.
4. Menjalankan perintah make clean untuk menghapus hasil kompilasi lama, kemudian make untuk membangun sistem yang baru.
5. Menjalankan sudo minicom dan melakukan reset pada VM Backend untuk melihat hasil output akhirnya."